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CH13: EARNED VALUE METHOD AND RELATED CONCEPTS

Prof Corne Schutte

Jou gunsteling vak?

Your favourite module?



- Kies jou geliefde nommer tussen 1 en 8
- Maal jou keuse met 3
- Tel 3 by
- Maal weer met 3
- Die antwoord behoort uit twee syfers te wees, bv 42
- Tel die twee syfers bymekaar, bv $4 + 2 = 6$
- Gebruik nou jou antwoord om op volgende skerm vas te stel wat is jou gunsteling vak.

- *Select your favourite number between 1 & 8*
- *Multiply your choice with 3*
- *Add 3*
- *Multiply again with 3*
- *The answer should consist of 2 digits, eg 42*
- *Add the two digits together, eg $4 + 2 = 6$*
- *Use your answer to select your favourite module on the next screen.*

Jou gunsteling vak?

1. Chemie / *Chemistry*
2. Fisika / *Physics*
3. Engels / *English*
4. Wiskunde / *Mathematics*
5. Termodinamika / *Thermodynamics*
6. Tegniese Tekeninge / *Technical Drawings*
7. Beheerstelsels / *Control Systems*
8. Ontwerp of “Skripsie” / *Design or Final Year Project*
9. Projekbestuur / *Project Management*
10. Toegepaste Wiskunde / *Applied Mathematics*

Plan!

- 13.5 (11.7-11.8*) – *Earned Value Analysis, you need it for project assignment 3.*
- *Think how you would apply it to your Project*
- *Plan:*
 - *Tuesday (7th) and Thursday (9th): Theory in 5 parts*
 - *Thursday 9th : Discuss EVA on Project (May not be required)*
 - *Friday 10th April: Quiz*
 - *Friday 10th April: Assignment 3 EVA Detail*

*Note difference in old and new text books.

4th Ed: 11.7 and 11.8.

5th Ed: All in 11.7

6th Ed: All in 13.6

7th Ed: All in 13.5

The Following Parts

- Part 0 – The context and necessity
- Part 1 – Illustrative Example 1 – how to calculate AC, PV and EV
- Part 2 – Illustrative Example 2 – expanding to SPI and CPI
- Part 3 – EVA Tracking
- Part 4 – A2/A3 Example
- Part 5 – Assignment 3 Context

Part 0: EVA Context

13.5: Performance Analysis and Earned Value Management



Percent Complete Concept

- *Subjective assessment of performance vs. schedule*
- *Assessed regularly for every work package*
- *Requires competency, candor, and trust*

13.5: *Earned Value*



Important

- *Performance Assessment*
 - *Purpose: determine **current status** of project with respect to **schedule** and **budget***
- *Earned value method*
 - *Combine the following information:*
 - *% complete,*
 - *schedule, and*
 - *cost*

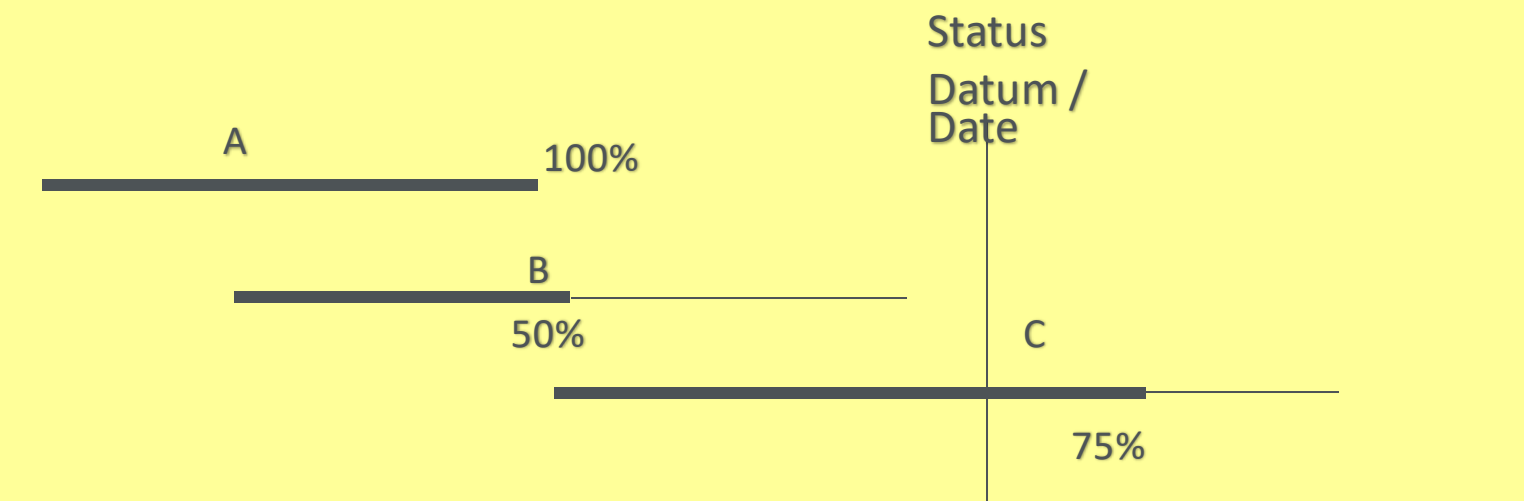


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Verdiende Waarde Konsep / Earned Value Concept (EV)

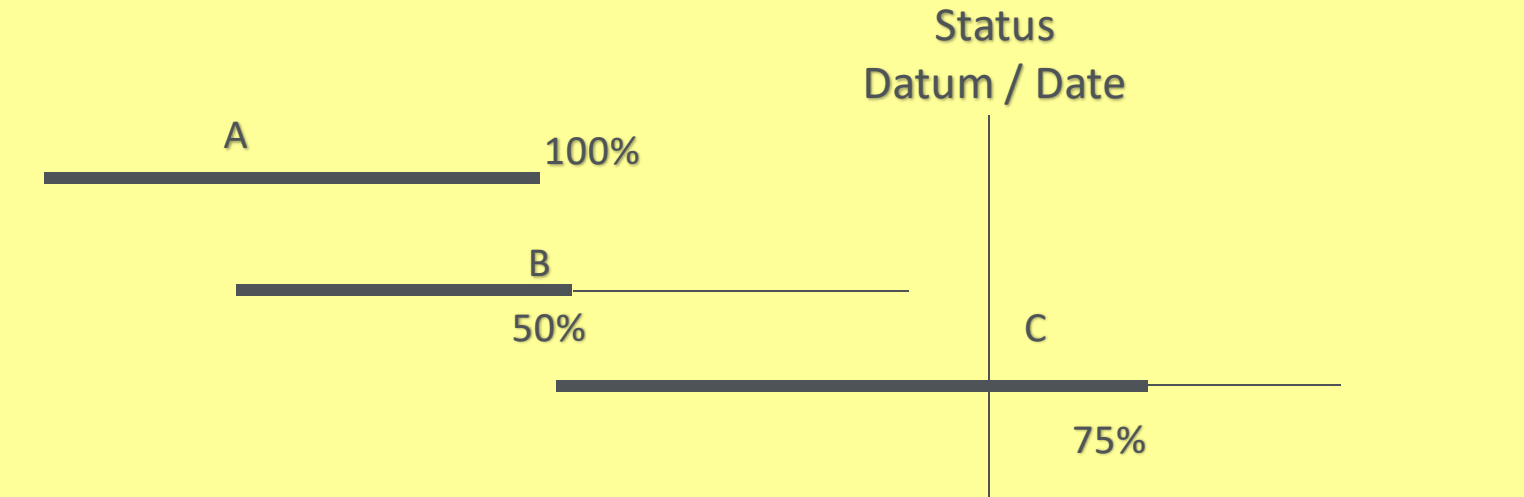


Taak / Task	Begroting / Budget	Uitgawes / Expenditure
A	1000	1000
B	2000	2050
C	1000	250
Tot	4000	3300

- A op skedule, op begroting /
A on schedule, on budget
 - B agter skedule, oor begroting /
B behind schedule, over budget
 - C voor skedule, onder begroting /
C ahead of schedule, under budget
- Hoe gaan dit met hierdie **projek**?
*How well is the **project** doing?*

Part 1: Example 1 - EVA Basics

Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC
	a	b
A	1000	1000
B	2000	2050
C	1000	250
Tot	4000	3300

Gegee van vorige bladsy / Given from previous page:

Begroting / Budget: Oorspronklike beplande koste – Basislyn / *Original planned cost – Baseline*

AC: Werklike koste sover spandeer / *Actual Cost spent*

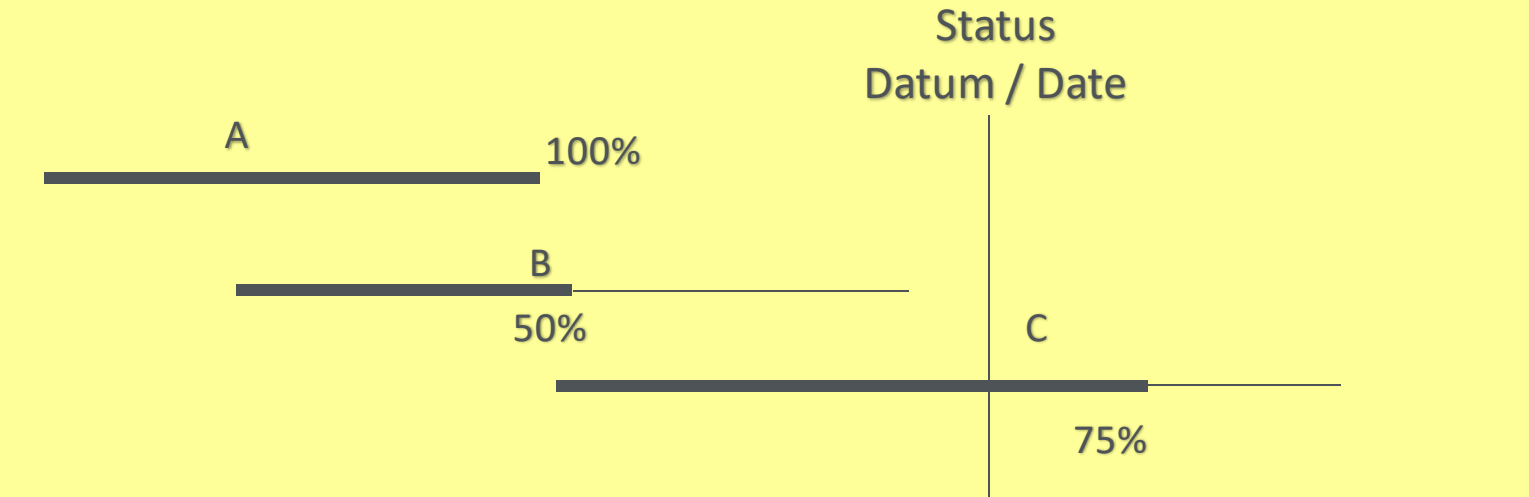


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Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC	% Sked / Sched
	a	b	c
A	1000	1000	100%
B	2000	2050	100%
C	1000	250	50%
Tot	4000	3300	

Bepaal kolom c / *Determine column c:*

% Skeduleer: Wat behoort die vordering te gewees het volgens my basislyn op die status datum?

% *Scheduled*: What should have been my progress according to my baseline on the status date?

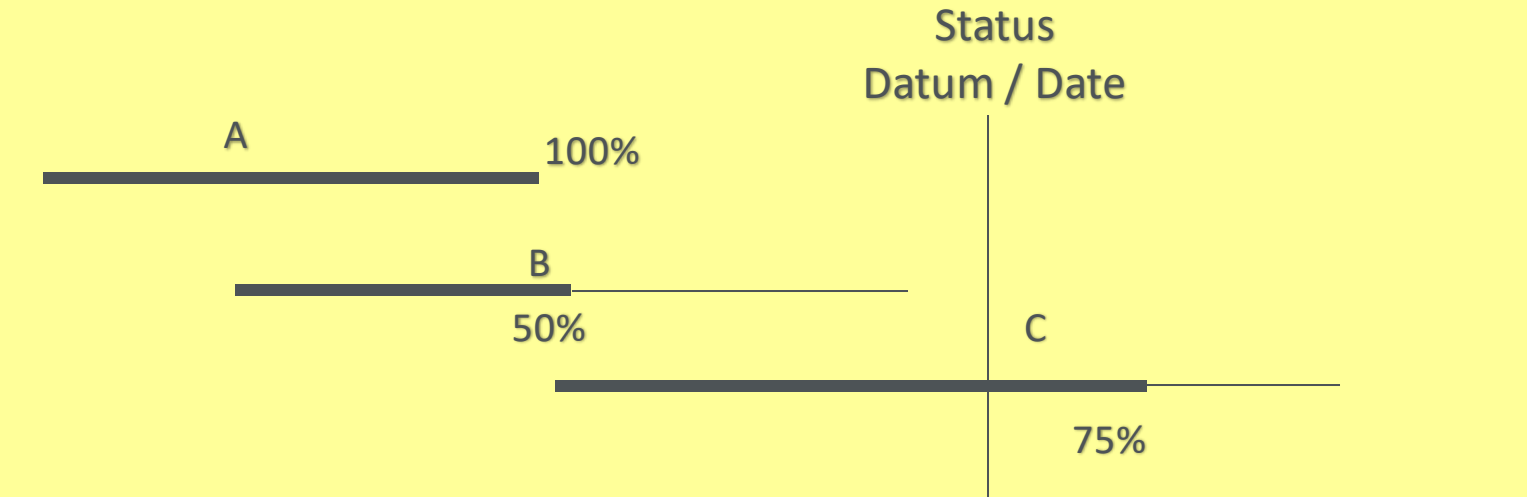


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Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC	% Sked / Sched	PV
	a	b	c	$a \times c$
A	1000	1000	100%	1000
B	2000	2050	100%	2000
C	1000	250	50%	500
Tot	4000	3300		3500

Bepaal Beplande Waarde / Determine PV:

Wat se waarde moes ek volgens my basislyn afgelewer het?

What value were I supposed to have delivered?

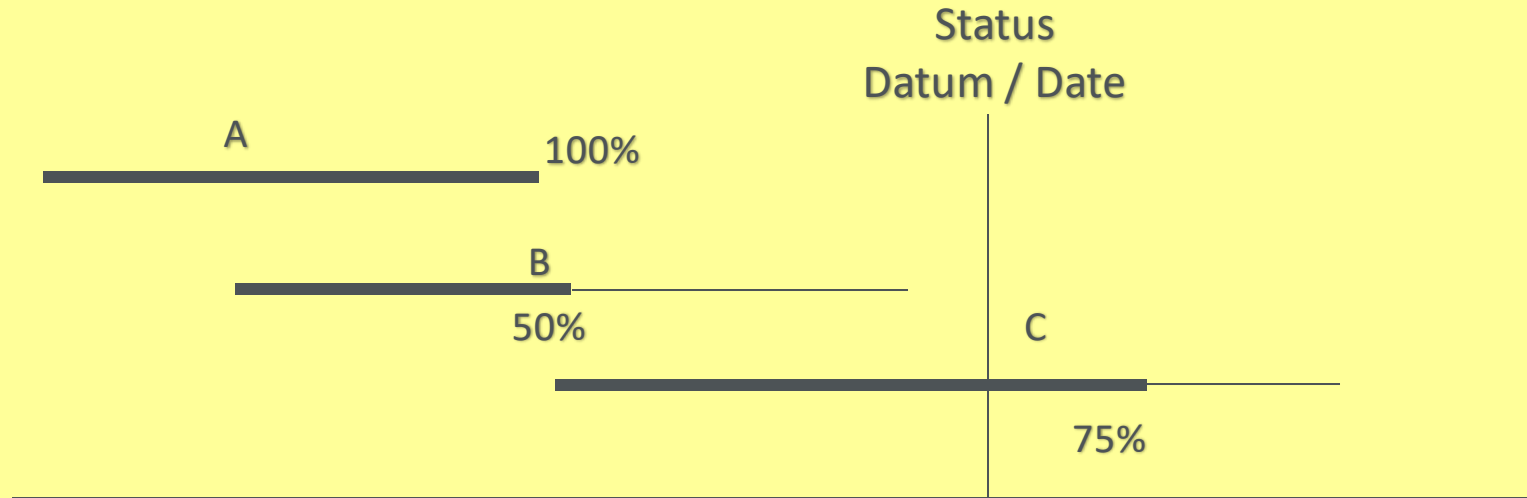


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Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC	% Sked / Sched	PV	% Voltooi / Cpl
	a	b	c	a x c	d
A	1000	1000	100%	1000	100%
B	2000	2050	100%	2000	50%
C	1000	250	50%	500	75%
Tot	4000	3300		3500	

Bepaal kolom d / Determine column d:

% Voltooi: Wat is my werklke vordering op die status datum?

% Complete: What is my real progression on the status date?

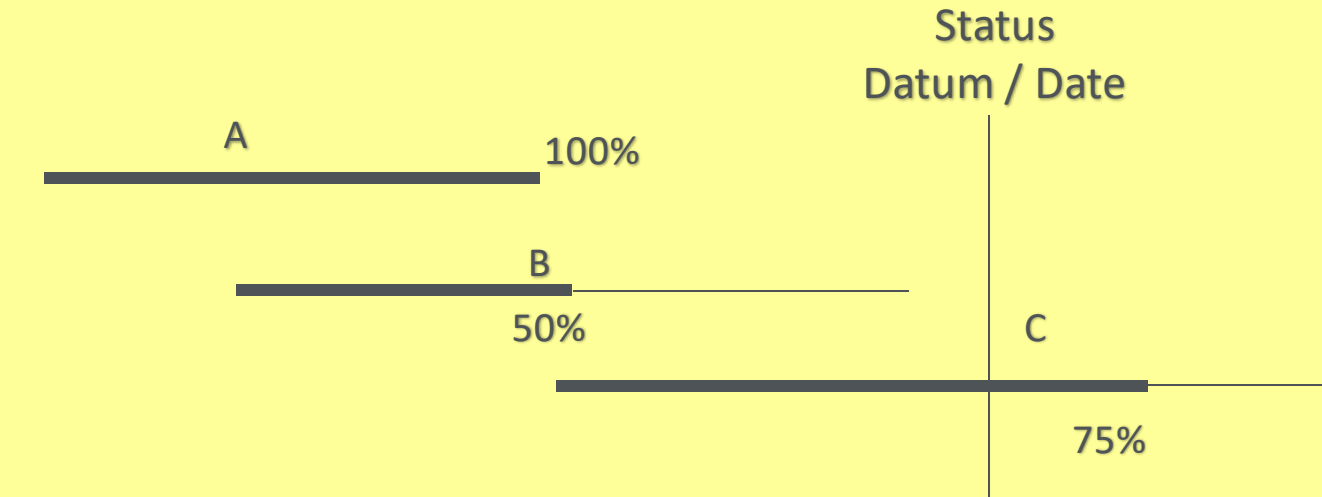


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Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC	% Sked / Sched	PV	% Voltooi / Cpl	EV
	a				d	a x d
A	1000				100%	1000
B	2000				50%	1000
C	1000				75%	750
Tot	4000	3300		3500		2750

Bepaal Verdiende Waarde (EV) / Determine Earned Value (EV):

EV: Bereken die waarde verdien op status datum
= a x d

EV: Calculate Earned Value on status date = a x d

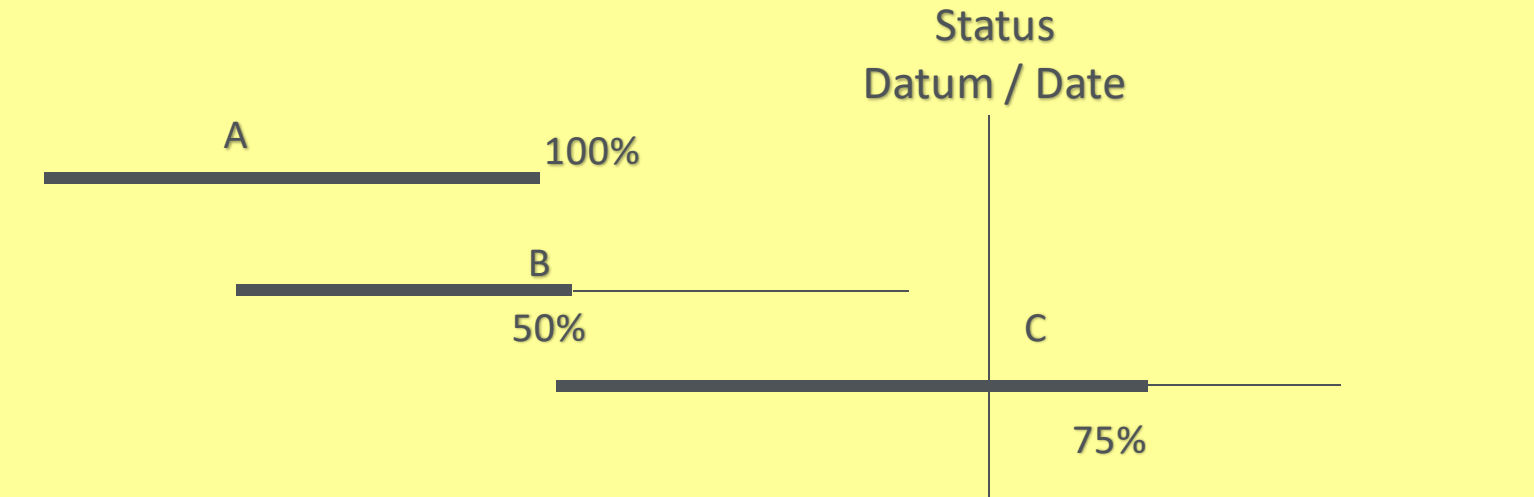


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Verdiende Waarde Konsep / Earned Value Concept (EV)



Taak / Task	Begroting / Budget	AC	% Sked / Sched	PV	% Voltooi / Cpl	EV
	a	b	c	a x c	d	a x d
A	1000	1000	100%	1000	100%	1000
B	2000	2050	100%	2000	50%	1000
C	1000	250	50%	500	75%	750
Tot	4000	3300		3500		2750

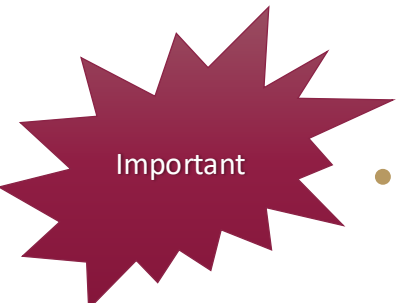


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Earned Value Definitions

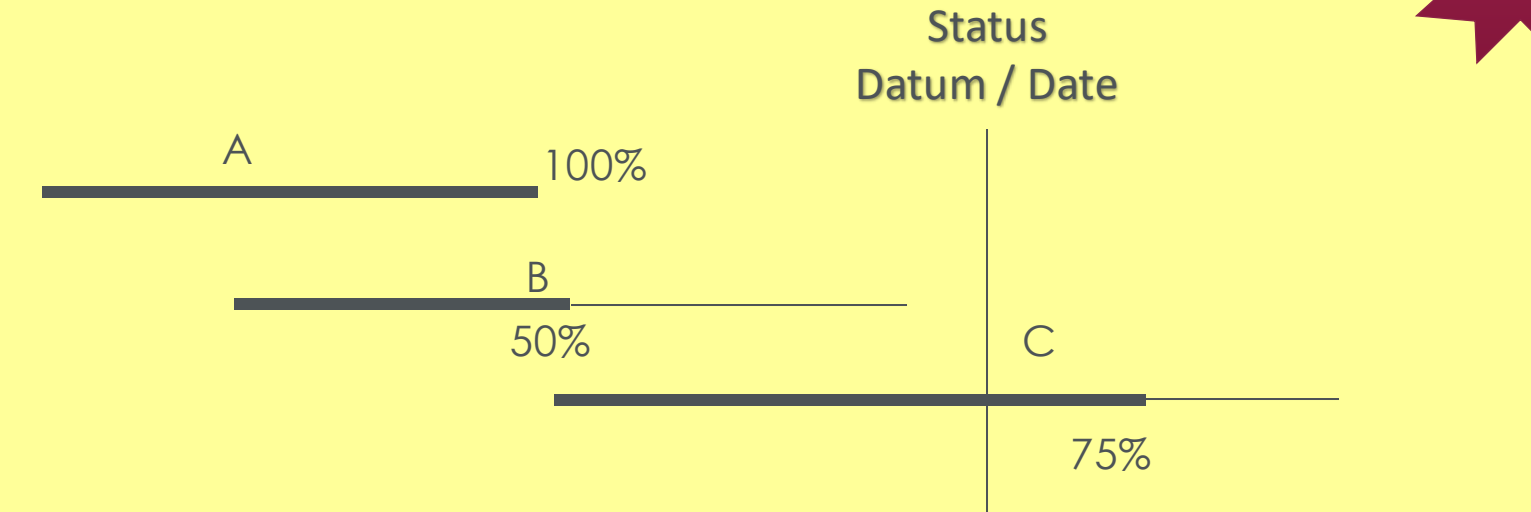


Important

- Planned Value / Beplande Waarde
 - PV = **p**lanned **v**alue (also called BCWS: budgeted cost of work scheduled)
- Actual Cost / Werklike Koste
 - AC = **a**ctual **c**ost (or ACWP: actual cost of work performed)
- Earned Value / Verdiende Waarde
 - EV = **e**arned **v**alue (or BCWP: budgeted cost of work performed)

Verdiende Waarde Konsep / Earned Value Concept (EV)

Important



Begroting / Budget	AC	PV	EV
4000	3300	3500	2750

- $EV < PV$:
Projek is agter skedule mbt waarde vd werk afgelewer (ignoreer kritieke pad)
*Project is **behind schedule** in terms of value of work (ignores critical path)*
- $AC > EV$
Projek is oor begroting in terme van waarde vd werk afgelewer.
*Project is **over budget** in terms of value of work completed*



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Part 2: Example 2 - Forecasting

13.5 Forecasting “To Complete” and “At Completion”



- *Second example*
 - *Assess not only current state of the project but also the project's likely completion cost.*

13.5: Voorbeeld / Example

Definities / Definitions

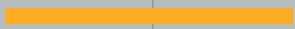
- $CPI = \text{cost performance index} = EV / AC$
- $SPI = \text{schedule performance index} = EV / PV$
- $ETC = \text{estimated cost to complete}$
 $= (BAC - EV) / CPI$
 $BAC = \text{budgeted cost at completion}$
- $EAC = \text{estimated cost at completion}$
 $= AC + ETC$



Important

13.5: Voorbeeld / Example



			Budget	% Complete	Cost Spent
A			A 220	100	240
B			B 190	100	180
C			C 250	50	150
D			D 90	100	110
E			E 350	50	190
F			F 400	100	380
G			G 250	0	0
H			H 140	0	0
I			I 240	0	0
	1 2 3 4	5 6 7 weeks	2,130		1,250

Today

Starting Point:

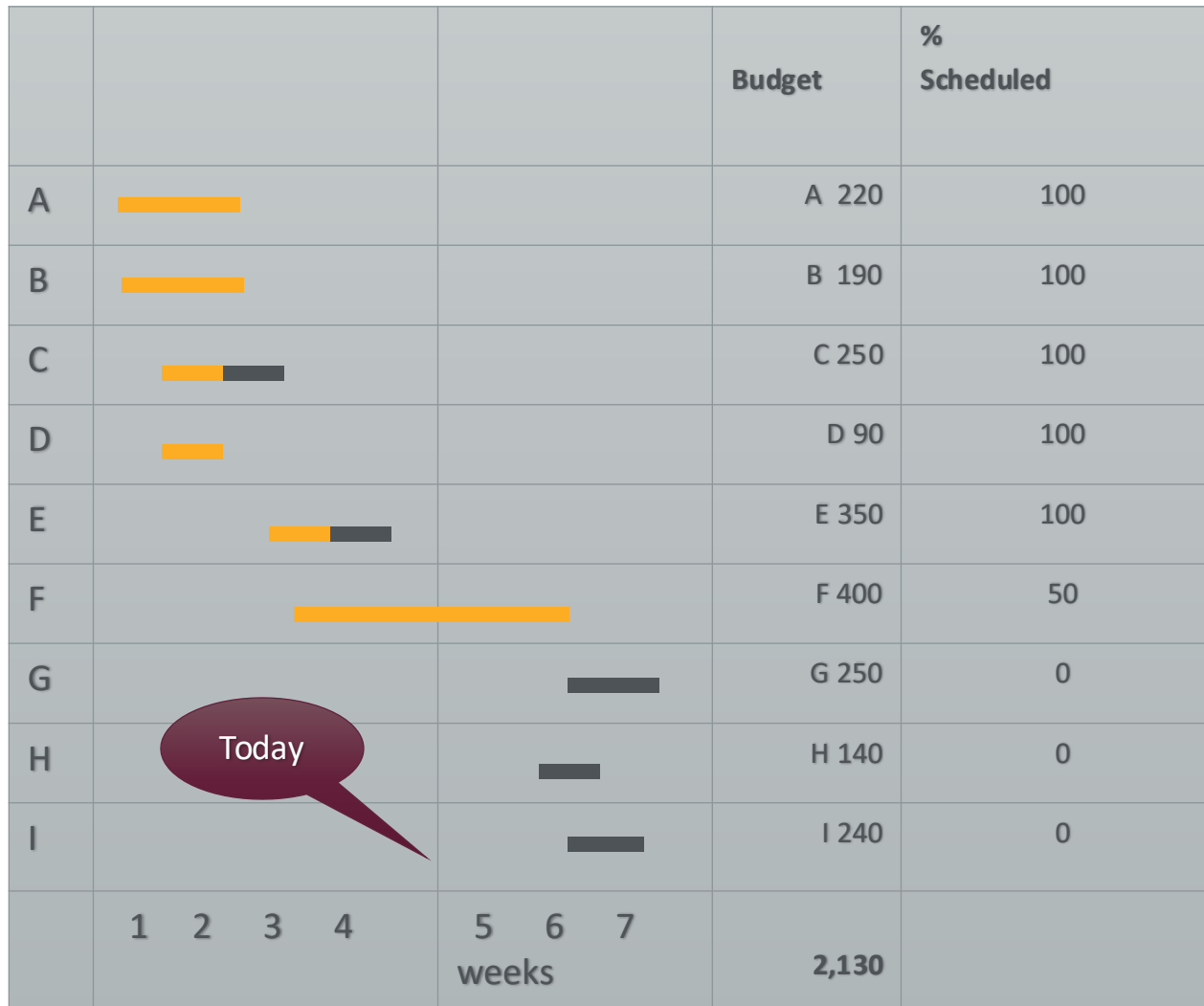
This is what is typically known “today”:

- What was my budget
- What is the current % completion
- How much have I spent to date.

In the example:

- We are at the end of week 4
- Budget for total project is 2130
- Had spent 1250 by end of week 4

13.5: Voorbeeld / Example



Calculate % Scheduled:

Where am I suppose to be at the status date?

Once I have % Sched, I can calculate my Planned Value – ie what value am I supposed to have delivered.

13.6: Voorbeeld / Example



			Budget	% Scheduled	PV
A			A 220	100	220
B			B 190	100	190
C			C 250	100	250
D			D 90	100	90
E			E 350	100	350
F			F 400	50	200
G			G 250	0	0
H			H 140	0	0
I			I 240	0	0
	1 2 3 4	5 6 7 weeks	2,130		1,300

Today

Calculate Planned Value:

$PV = \% \text{ Sched} \times \text{Budget}$
 (This is the value I should have delivered by “today”)

13.5: Voorbeeld / Example



			Budget	% Scheduled	PV	% Complete	EV
A			A 220	100	220	100	220
B			B 190	100	190	100	190
C			C 250	100	250	50	125
D			D 90	100	90	100	90
E			E 350	100	350	50	175
F			F 400	50	200	100	400
G			G 250	0	0	0	0
H			H 140	0	0	0	0
I			I 240	0	0	0	0
	1 2 3 4	5 6 7 weeks	2,130		1,300		1,200

Today

Calculate Earned Value:

$EV = (\% Cpl) \times \text{Budget}$

This what I actually delivered by “today”

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7							
		weeks	2,130		1,300		1,200	1,250	

Week 4, **Project Cost to date:**

– AC= \$1,250



13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7							
		weeks	2,130		1,300		1,200	1,250	

Week 4:

Value of work completed:

– EV = \$1,200



13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7 weeks	2,130		1,300		1,200	1,250	

Week 4: / *Value of work remaining:*

$$- \text{BAC} - \text{EV} = \$2,130 - \$1,200 = \$930$$

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7 weeks	2,130		1,300		1,200	1,250	

Week 4: **Cost Performance Index (CPI)**

– $EV/AC = 1,200/1,250 = 0.96$

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7	2,130		1,300		1,200	1,250	
		weeks							

Week 4: **Schedule Performance Index (SPI)**

– $EV / PV = 1200 / 1300 = 0.92$

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7 weeks	2,130		1,300		1,200	1,250	

- **Likely remaining cost to complete project (ETC)**
 - $(BAC - EV) / CPI = (2130 - 1200) / 0.96 = \969

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7 weeks	2,130		1,300		1,200	1,250	

- **Likely project cost at completion**
 - $EAC = AC + ETC = \$1,250 + \$969 = \$2,219$

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 90	100	90	100	90	110	
E			E 350	100	350	50	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7	2,130		1,300		1,200	1,250	
		weeks							

- **Project cost variance**

- $BAC - EAC = \$2,130 - \$2,219 = -\$89$ (4.2% overrun)

13.5: Voorbeeld / Example



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			Budget	% Scheduled	PV	% Complete	EV	AC	
A			A 220	100	220	100	220	240	
B			B 190	100	190	100	190	180	
C			C 250	100	250	50	125	150	
D			D 80	100	80	100	90	110	
E			E 350	100	350	100	175	190	
F			F 400	50	200	100	400	380	
G			G 250	0	0	0	0	0	
H			H 140	0	0	0	0	0	
I			I 240	0	0	0	0	0	
	1 2 3 4	5 6 7							
		weeks	2,130		1,300		1,200	1,250	

Context of your S4P project assignment 3

Do you know where to get all these figures? Think about it...

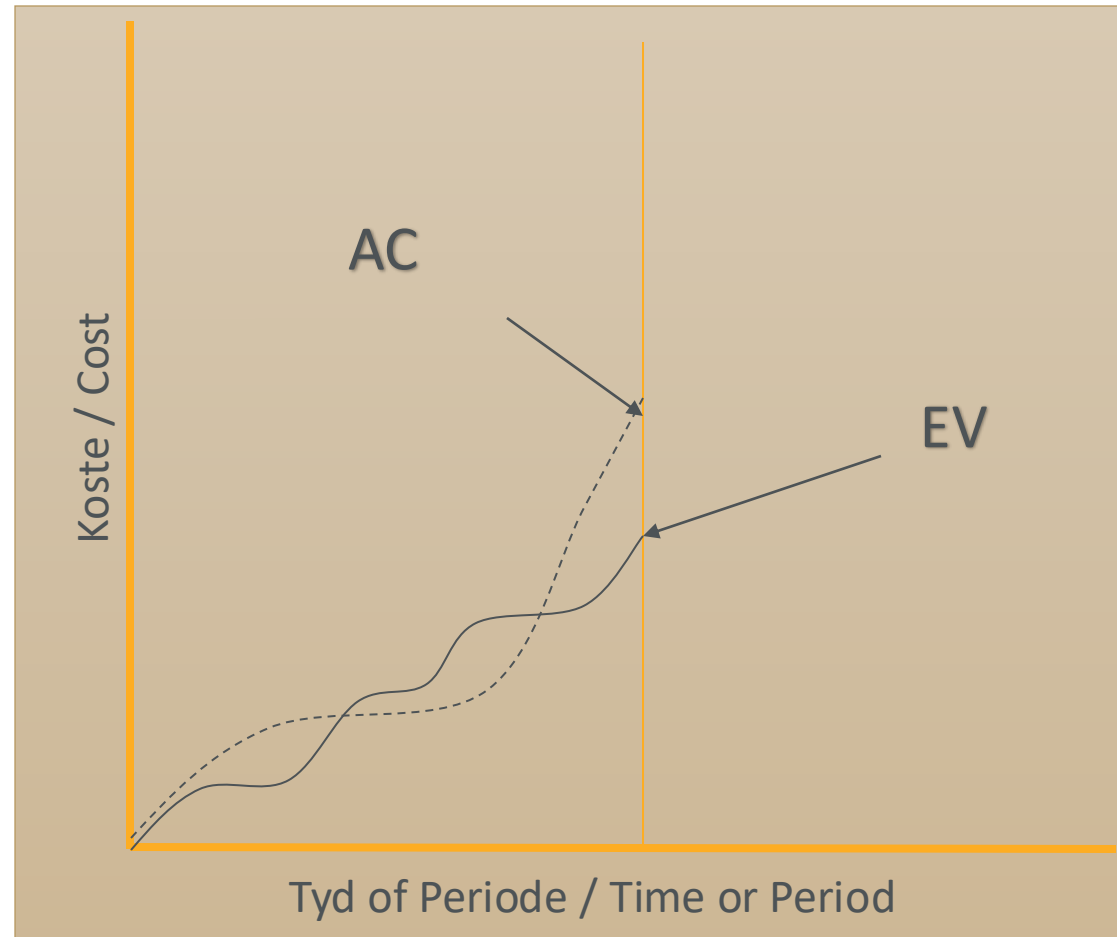
Part 3: Tracking

13.5: Verdiende Waarde Nasporing / *Earned Value Tracking*

- *Earned value is assessed continually*
- *Actual expenditures (AC) and actual work completed (EV) are assessed vs. budget and schedule (PV)*

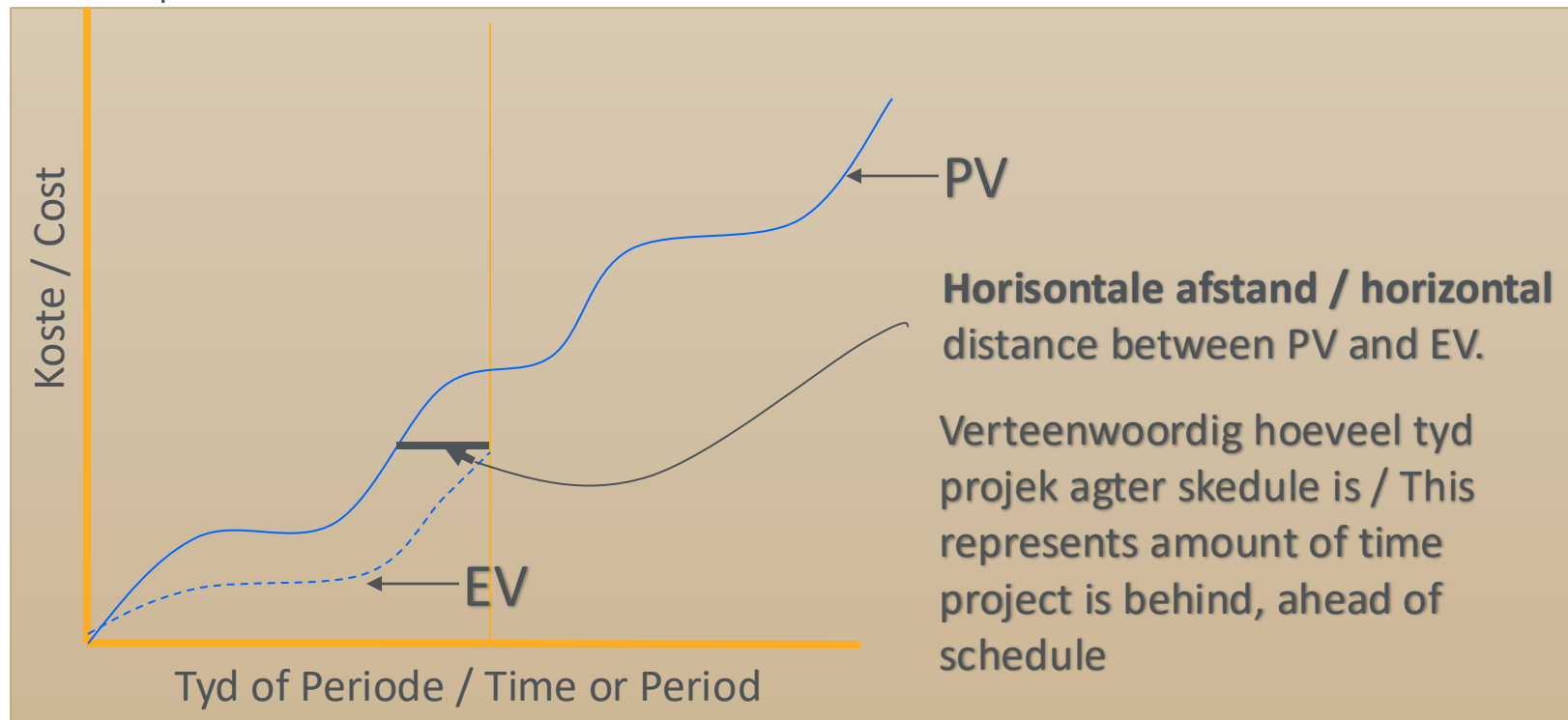
13.5: Verdiende Waarde Nasporing / *Earned Value Tracking*

- If $AC > EV$ = project is over budget



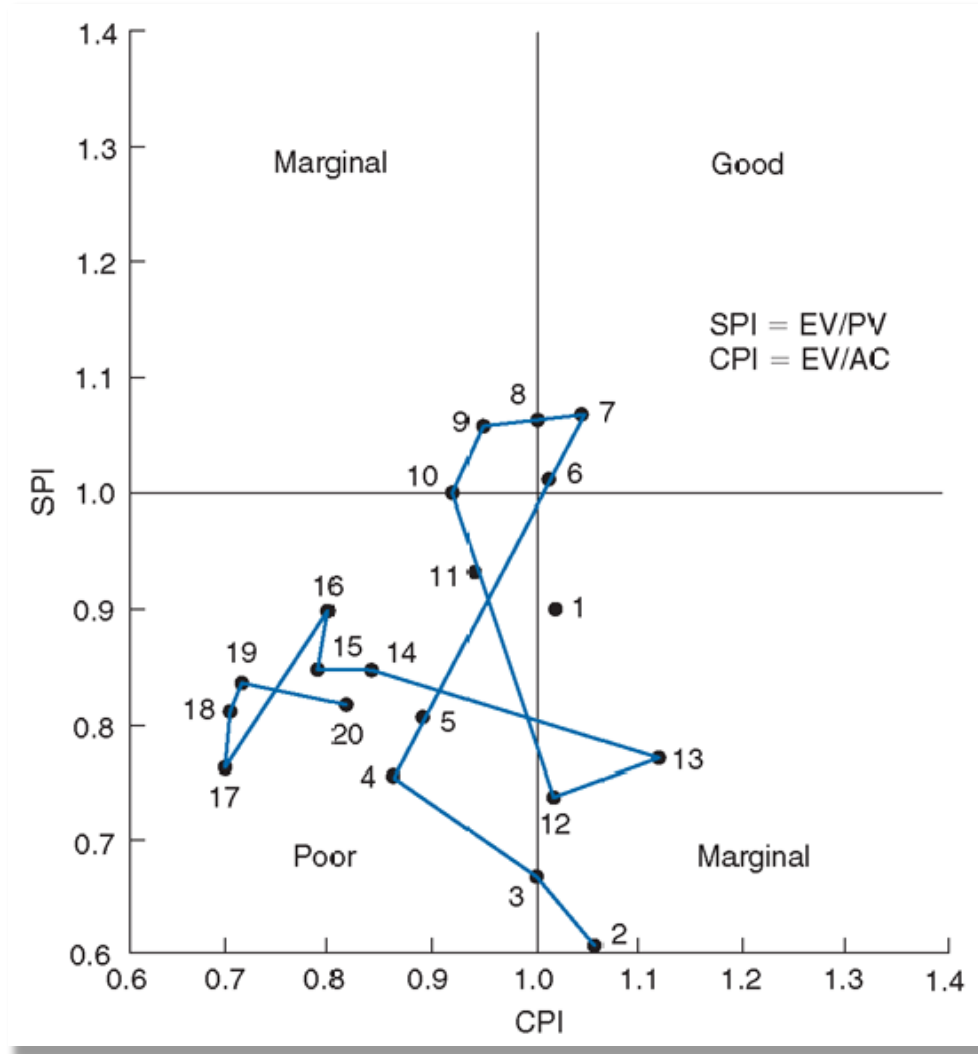
13.5: Verdiende Waarde Nasporing / *Earned Value Tracking*

- *If $EV < PV$ = project is behind schedule*
- *How far behind schedule?*
 - Progress on the critical path, or
 - Compare horizontal distance tussen EV with PV



13.6: Verdiende Waarde Nasporing / *Earned Value Tracking*

LOGON project cost / schedule performance plotted for months 1-20



Waarde van Verdiende Waarde / *Value of Earned Value ?*



Important

- *Enables*
 - *Simultaneous assessment of % completion, schedule, and cost performance*
 - *Information about work packages to be rolled up to provide a project-wide assessment*
 - *Forecasts of cost and date at completion*
- *Assumptions*
 - *Valid estimates of % complete*
 - *Actual expenditures coincide with work performed (no delayed or advance payment)*

Addisionele materiaal / Additional Material



- IIE Publication:
 - [Earned Value Analysis Example](#)
- PMI Standard:
 - [Practice Standard for Earned Value Management](#)

(Sien SUNLearn / See SUNLearn)

Part 4: A2/A3 Example

2015 A2 Example



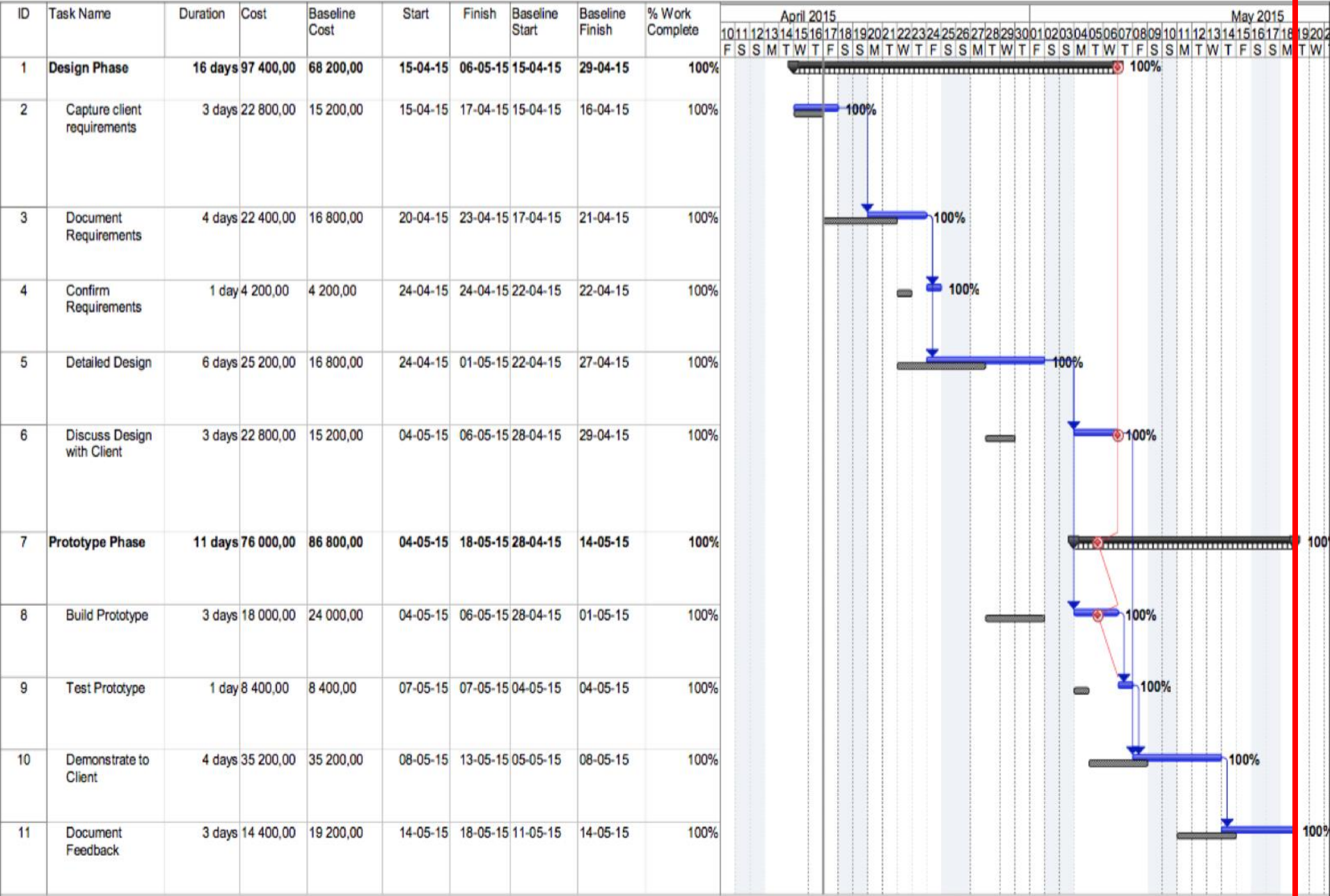
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Vraag 7: Verdienende Waarde Analise	33	Question 7: Earned Value Analysis
Beskou die projekstatus van die MS Project plan (sien bylaag). Die analisedatum van die gegewe plan is op die einde van die dag wat die projek voltooid is (dit is gistermiddag 18 Mei 2015). Aannames en notas: - Om berekeninge te vereenvoudig, neem aan daar was geen publieke vakansiedae gedurende die periode nie. - Daar was nie gedurende naweke gewerk nie. - Vordering en kostes vermeerder teen 'n konstante tempo vir elke taak. - Wys altyd die formule wat gebruik was vir elke berekening. Die onderstaande vrae se berekeninge moet vir die volgende datum gemaak word: Datum 1: Die datum waarop die projek voltooi is (maw einde van die dag op 18 Mei - dit is gistermiddag)		<i>Consider the project status of the MS Project Plan (see appendix). The analysis date of the given plan is close of business on the day the project was completed – (that is yesterday afternoon the 18th of May 2015).</i> Assumptions and notes: <i>To simplify calculations, assume there were no public holidays during this period.</i> <i>No work occurred during weekends.</i> <i>Progress and costs accrue at a constant rate for each task.</i> <i>Always show your formula used for each calculation.</i> <i>The calculations in the questions below must be made for the following date:</i> Date 1: <i>The date on which the project was completed (ie close of business on 18 May – yesterday afternoon)</i>

MS Project Plan



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2015 A2 Example

a. Deur net die MSProject Plan uitdruk te bestudeer sonder enige verdiende waarde berekeninge, wat is jou indrukke mbt die projek se verrigting in terme van begroting en skedule? Motiveer.	2	a. <i>By only studying the MSProject Plan printout, without performing any EVA calculations, what are your impressions wrt this project's performance in terms of budget and schedule? Explain.</i>
b. Waarom word geen kritieke pad take op die MSProject Plan uitdruk aangedui nie?	1	b. <i>Why are there no critical path tasks indicated on the MSProject Plan printout?</i>

Memo:

- a) *BL finish is before Finish, project thus **behind schedule**. Cost is greater than Baseline cost, thus **over budget**.*
- b) *MS Project plan's status date is on completed project, so no critical tasks left to do.*



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2015 A2 Example



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Analises op Datum 1:		<i>Date 1 Analysis:</i>
c. Verskaf 'n tabel wat die berekeninge bevat vir die totale projek op datum 1 , vir elke taak en taak groepering, van die werklike koste (AC), beplande waarde (PV), en verdiende waarde (EV). (Toon die formules gebruik vir elke kolom)	6	c. <i>Provide a table that contains the calculation of the total project on date 1, for each task and task group, of the actual cost (AC), planned value (PV), and earned value (EV). (Show the formulas used for each column)</i>
d. Bereken die Kosteverrigtingsindikator (CPI) op datum 1 .	2	d. <i>Calculate the Cost Performance Indicator (CPI) on date 1.</i>
e. Bereken die skeduleverrigtingsindikator (SPI) op datum 1 .	2	e. <i>Calculate the Schedule Performance Indicator (SPI) on date 1.</i>
f. Gegee jou analises hier bo, bespreek die projek se verrigting en verdiende waarde analise (EVA) se toepaslikheid mbt die berekenings met voltooiing van die projek (datum 1).	4	f. <i>Given your analyses above, discuss the project's performance and the applicability of Earned Value Analysis calculations at completion of the project (date 1).</i>

Memo 2



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Evaluation Date (COB)	18-May (end of day)											
Task Name	Duration	Cost	Baseline Cost	Start	Finish	Baseline Start	Baseline Finish	% Complete	Sch %	c. AC	c. PV	c. EV
Design Phase	16 days	97 400,00	68 200,00	15-04-2015	06-05-2015	15-04-2015	29-04-2015			R 97 400,00	R 68 200,00	R 68 200,00
Capture client requirements	3 days	22 800,00	15 200,00	15-04-2015	17-04-2015	15-04-2015	16-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Document Requirements	4 days	22 400,00	16 800,00	20-04-2015	23-04-2015	17-04-2015	21-04-2015	100%	100%	R 22 400,00	R 16 800,00	R 16 800,00
Confirm Requirements	1 day	4 200,00	4 200,00	24-04-2015	24-04-2015	22-04-2015	22-04-2015	100%	100%	R 4 200,00	R 4 200,00	R 4 200,00
Detailed Design	6 days	25 200,00	16 800,00	24-04-2015	01-05-2015	22-04-2015	27-04-2015	100%	100%	R 25 200,00	R 16 800,00	R 16 800,00
Discuss Design with Client	3 days	22 800,00	15 200,00	04-05-2015	06-05-2015	28-04-2015	29-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Prototype Phase	11 days	76 000,00	86 800,00	04-05-2015	18-05-2015	28-04-2015	14-05-2015			R 76 000,00	R 86 800,00	R 86 800,00
Build Prototype	3 days	18 000,00	24 000,00	04-05-2015	06-05-2015	28-04-2015	01-05-2015	100%	100%	R 18 000,00	R 24 000,00	R 24 000,00
Test Prototype	1 day	8 400,00	8 400,00	07-05-2015	07-05-2015	04-05-2015	04-05-2015	100%	100%	R 8 400,00	R 8 400,00	R 8 400,00
Demonstrate to Client	4 days	35 200,00	35 200,00	08-05-2015	13-05-2015	05-05-2015	08-05-2015	100%	100%	R 35 200,00	R 35 200,00	R 35 200,00
Document Feedback	3 days	14 400,00	19 200,00	14-05-2015	18-05-2015	11-05-2015	14-05-2015	100%	100%	R 14 400,00	R 19 200,00	R 19 200,00
		R 173 400,00	R 155 000,00							R 173 400,00	R 155 000,00	R 155 000,00
Value of rem work (BAC-EV)	R 0,00											
d. CPI	0,894											
e. SPI	1,000											
ETC	R 0,00											
EAC	R 173 400,00											

- Note left side (in black) is exact copy of MS Project Plan table
- Red figures are what have been calculated

Memo 2 zoomed

[illegible]

Memo 2 -the SPI mystery



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Evaluation Date (COB)	18-May (end of day)											
Task Name	Duration	Cost	Baseline Cost	Start	Finish	Baseline Start	Baseline Finish	% Complete	Sch %	c. AC	c. PV	c. EV
Design Phase	16 days	97 400,00	68 200,00	15-04-2015	06-05-2015	15-04-2015	29-04-2015			R 97 400,00	R 68 200,00	R 68 200,00
Capture client requirements	3 days	22 800,00	15 200,00	15-04-2015	17-04-2015	15-04-2015	16-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Document Requirements	4 days	22 400,00	16 800,00	20-04-2015	23-04-2015	17-04-2015	21-04-2015	100%	100%	R 22 400,00	R 16 800,00	R 16 800,00
Confirm Requirements	1 day	4 200,00	4 200,00	24-04-2015	24-04-2015	22-04-2015	22-04-2015	100%	100%	R 4 200,00	R 4 200,00	R 4 200,00
Detailed Design	6 days	25 200,00	16 800,00	24-04-2015	01-05-2015	22-04-2015	27-04-2015	100%	100%	R 25 200,00	R 16 800,00	R 16 800,00
Discuss Design with Client	3 days	22 800,00	15 200,00	04-05-2015	06-05-2015	28-04-2015	29-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Prototype Phase	11 days	76 000,00	86 800,00	04-05-2015	18-05-2015	28-04-2015	14-05-2015			R 76 000,00	R 86 800,00	R 86 800,00
Build Prototype	3 days	18 000,00	24 000,00	04-05-2015	06-05-2015	28-04-2015	01-05-2015	100%	100%	R 18 000,00	R 24 000,00	R 24 000,00
Test Prototype	1 day	8 400,00	8 400,00	07-05-2015	07-05-2015	04-05-2015	04-05-2015	100%	100%	R 8 400,00	R 8 400,00	R 8 400,00
Demonstrate to Client	4 days	35 200,00	35 200,00	08-05-2015	13-05-2015	05-05-2015	08-05-2015	100%	100%	R 35 200,00	R 35 200,00	R 35 200,00
Document Feedback	3 days	14 400,00	19 200,00	14-05-2015	18-05-2015	11-05-2015	14-05-2015	100%	100%	R 14 400,00	R 19 200,00	R 19 200,00
		R 173 400,00	R 155 000,00							R 173 400,00	R 155 000,00	R 155 000,00
Value of rem work (BAC-EV)	R 0,00											
d. CPI	0,894											
e. SPI	1,000											
ETC	R 0,00											
EAC	R 173 400,00											



- SPI = 1? Why?
 - We know the project is behind schedule, so SPI should be less than 1?
 - For the answer, see the additional article on SUNLearn!
- ETC = 0? Why?
 - The project is finished, so nothing left to do. EAC is thus equal to AC.

2015 A2 Example

- Notes:
 - Note the hierarchy of tasks – do not sum twice! Only calculate on lowest level.
 - Why can't you just calculate on the summary tasks?
 - The plan shows status of end of project.
 - What would happen if I ask you to do another date?
 - Actual Cost calc? (Important: Where does the cost column in MSProject come from?)
 - % Scheduled?
 - % Actual?
 - Do now the same calculation on COB 6 May

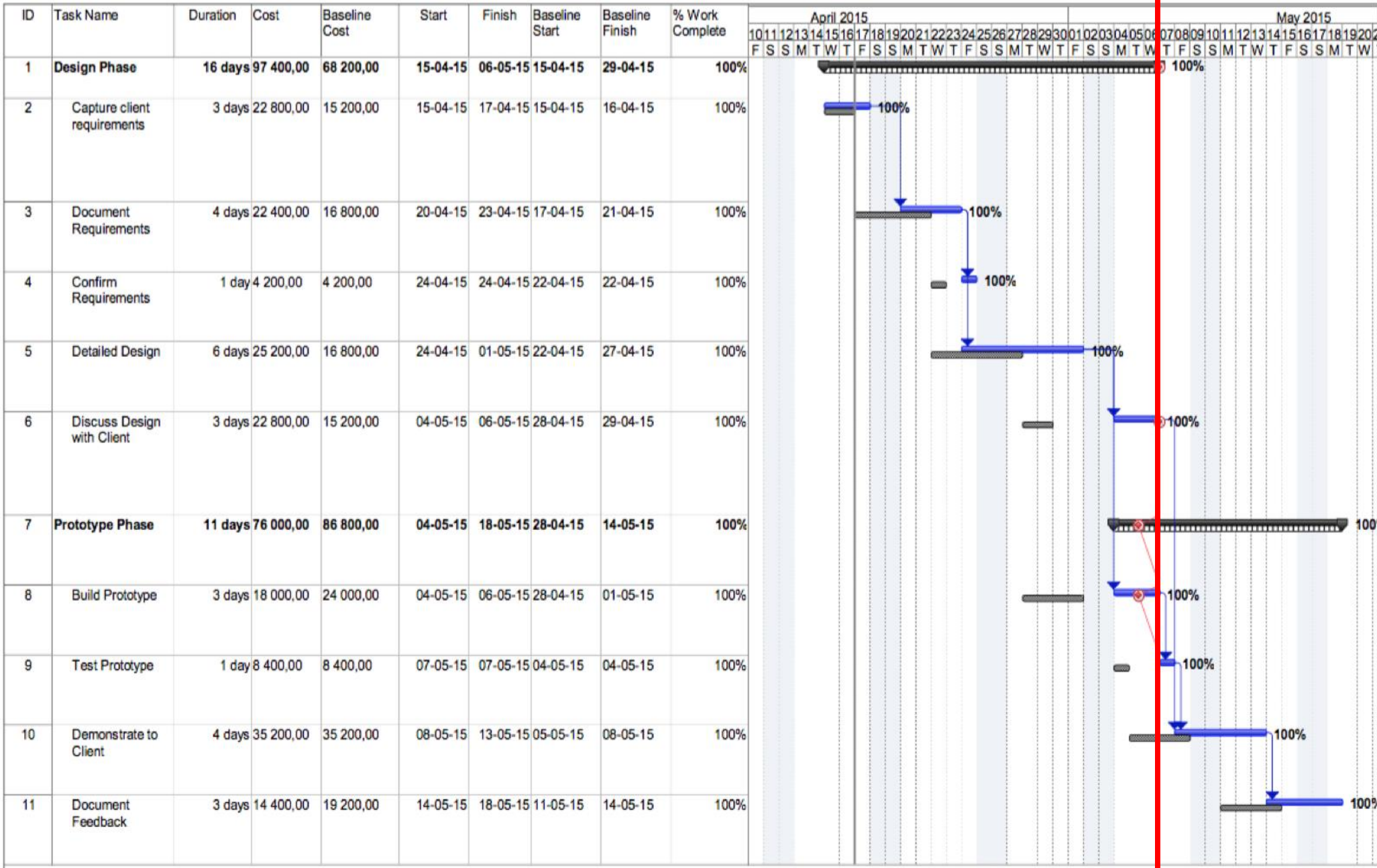
MS Project Plan with date 6 May



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Let us complicate it....

- With date COB 6 May

Evaluation Date (COB)	06-May (end of day)											
Task Name	Duration	Cost	Baseline Cost	Start	Finish	Baseline Start	Baseline Finish	% Complete	Sch %	g. AC	g. PV	g. EV
Design Phase	16 days	97 400,00	68 200,00	15-04-2015	06-05-2015	15-04-2015	29-04-2015			R 97 400,00	R 68 200,00	R 68 200,00
Capture client requirements	3 days	22 800,00	15 200,00	15-04-2015	17-04-2015	15-04-2015	16-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Document Requirements	4 days	22 400,00	16 800,00	20-04-2015	23-04-2015	17-04-2015	21-04-2015	100%	100%	R 22 400,00	R 16 800,00	R 16 800,00
Confirm Requirements	1 day	4 200,00	4 200,00	24-04-2015	24-04-2015	22-04-2015	22-04-2015	100%	100%	R 4 200,00	R 4 200,00	R 4 200,00
Detailed Design	6 days	25 200,00	16 800,00	24-04-2015	01-05-2015	22-04-2015	27-04-2015	100%	100%	R 25 200,00	R 16 800,00	R 16 800,00
Discuss Design with Client	3 days	22 800,00	15 200,00	04-05-2015	06-05-2015	28-04-2015	29-04-2015	100%	100%	R 22 800,00	R 15 200,00	R 15 200,00
Prototype Phase	11 days	76 000,00	86 800,00	04-05-2015	18-05-2015	28-04-2015	14-05-2015			R 18 000,00	R 50 000,00	R 24 000,00
Build Prototype	3 days	18 000,00	24 000,00	04-05-2015	06-05-2015	28-04-2015	01-05-2015	100%	100%	R 18 000,00	R 24 000,00	R 24 000,00
Test Prototype	1 day	8 400,00	8 400,00	07-05-2015	07-05-2015	04-05-2015	04-05-2015	0%	100%	R 0,00	R 8 400,00	R 0,00
Demonstrate to Client	4 days	35 200,00	35 200,00	08-05-2015	13-05-2015	05-05-2015	08-05-2015	0%	50%	R 0,00	R 17 600,00	R 0,00
Document Feedback	3 days	14 400,00	19 200,00	14-05-2015	18-05-2015	11-05-2015	14-05-2015	0%	0%	R 0,00	R 0,00	R 0,00
		R 173 400,00	R 155 000,00							R 115 400,00	R 118 200,00	R 92 200,00
Value of rem work (BAC-EV)	R 62 800,00											
h. CPI	0,799											
i. SPI	0,780											
ETC	R 78 602,17											
j. EAC	R 194 002,17											

- Be very careful now of the MS Project Plan info
 - The actual % completions must now be adjusted, and not just taken from the Gantt chart
 - The cost shown there is for 18 May, but you now need to calculate cost for 6 May – luckily here, the date of 6 May was nicely selected, but it may mean that you need to pro-rate Actual Cost.



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Let us complicate it... zoomed

- Calculation on COB 6 May for example

[illegible]

Final Comments



- This is very typical of what you can expect in A2/A3.
- It is also a good exposure for Assignment 3, but the additional complexities will be discussed in the Tut.
- Actual Cost:
 - The Cost provided by MS Project is for status date 18 May.
 - You thus need to pro-rate AC to get to the AC of 6 May.
- Task durations (planned and actual) are full days – a simplification. This is not true for assignment 3.
- 6 May date was carefully selected to make problem easier:
 - All tasks either 0% or 100%
 - Make sure you can also do the calculation where the date is in the middle of a set of tasks.

Part 5: Assignment 3

Assignment 3 EVA Context



- Consists of:
 - Excel spreadsheet with EVA calculations
 - including a period 9 'progress report'
 - “Post Mortem” Review meeting
 - Project Closure Report (based on EVA calc)
 - Final Updated Project Plan
- Excel Spreadsheet:
 - Calculations for each period must be on a **separate sheet** in the Excel file (ie. 12 sheets)
 - On a separate **summary sheet** in the Excel file (make it the first sheet)
 - Period 9 'Snapshot' - as if giving a progress report at the end of period 9 (P9 min 4 workdays)
 - EVA Graphs + estimated cost at completion
 - Include on this **summary sheet** a short discussion on the status of your project **as at the end of period 9**, using the graphs at your disposal.

Progress Reports

- *Current status of project performance (traffic light).*
- *Accomplishments since prior report*
- *Red-flagged items*
 - *Progress toward resolution of problems.*
 - *Planned corrective actions.*
- *Schedule (changes, forecast etc)*
- *Milestones expected to be reached during next reporting period.*
- *Current and potential problems.*
- *Current Cost situation and cost performance*
- *Manpower plan and limitations*



Tip: Use the above as a guideline for the structure of the first sheet in Assignment 3

Assignment 3 EVA Details



- Include EVA:
 - Use as analysis dates, the close of business, **4 working days** before the **actual** completion date of each simulated period.
 - Each period must be on a **separate sheet** in the Excel file
 - On a separate **summary sheet** in the Excel file (make it the first sheet)
 - Calculation of AC, PV and EV, SPI & CPI for the **end** of Period 9 (COB min 4 workdays).
 - Calculation of **likely project cost at completion**, based on data up to the end of Period 9 (COB min 4 workdays).
 - Plot **AC and EV**, and **PV and EV** for all periods, and explain the results within the context of your simulated project's progress.
 - Plot **SPI vs CPI** for all periods, and discuss the trends seen, within the context of your own simulated project.
 - Include on this **summary sheet** a short discussion on the status of your project, discussing time and cost, using the graphs at your disposal.
 - Bonus points will be earned if you include **Training and Management** activities in your Earned Value Analysis.

Thank you
Enkosi
Dankie

